

6 Glucose is an example of a monosaccharide.

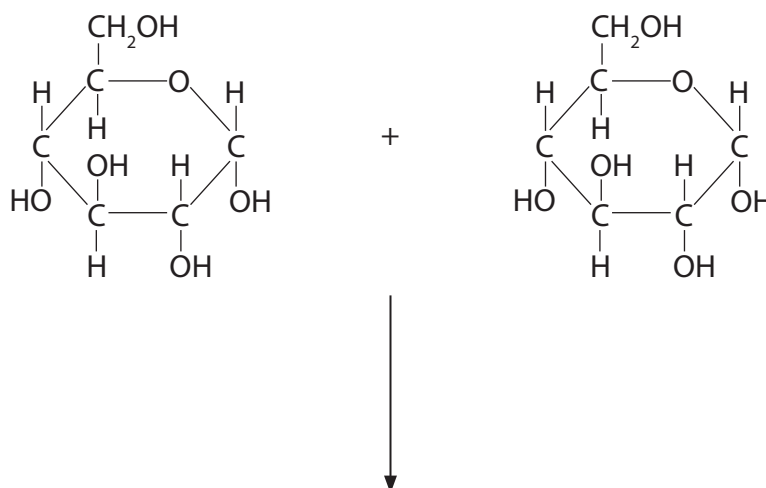
Monosaccharides join together to form disaccharides, oligosaccharides and polysaccharides.

Oligosaccharides contain from three to ten monosaccharides.

(a) The diagram shows two glucose molecules.

Complete the diagram to show how these two molecules join by a glycosidic bond to form **two** products.

(3)



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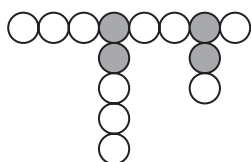
(b) Starch is a polysaccharide composed of amylose and amylopectin.

Digestion of starch begins in the mouth by an enzyme called amylase.

This amylase can break only 1–4 glycosidic bonds.

Only disaccharides and oligosaccharides are produced during this reaction.

(i) The diagram shows part of an amylopectin molecule.



Key

○ glucose molecule joined by 1–4 glycosidic bonds

● glucose molecule joined by 1–4 and 1–6 glycosidic bonds

Draw **one** disaccharide and **two** different oligosaccharides that could be produced following the digestion of this part of amylopectin by this amylase.

(2)

(ii) Explain why this amylase is able to break only 1–4 glycosidic bonds.

(2)

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(c) The digestion of carbohydrates is completed in the small intestine.

Glucose is then absorbed into the bloodstream.

- (i) Suggest how disaccharides and oligosaccharides are broken down in the small intestine.

Give reasons for your answer.

(3)

- (ii) The absorption of glucose into the bloodstream requires membrane transport proteins.

Explain why glucose cannot diffuse into the bloodstream.

(2)

(Total for Question 6 = 12 marks)

